

# On the Universality of Comparability Grids for Measurement-Based Quantum Computation

Ryan Y. Batubara

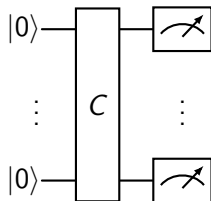
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UCSD

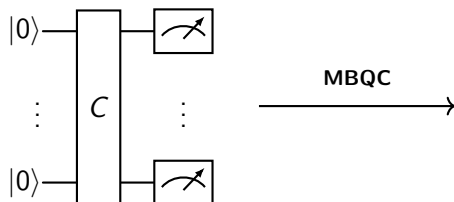
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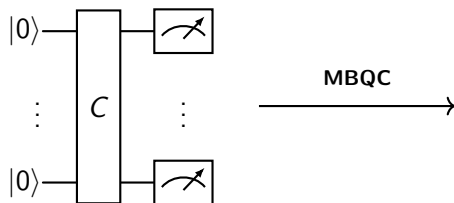
# Measurement-Based Quantum Computation (MBQC)



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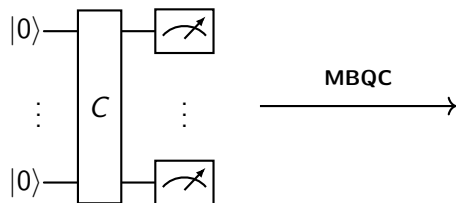


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- ▶ "resource state"

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- ▶ Single-qubit measurements on a
- ▶ "resource state"
- ▶ Excluding resource state, done in constant depth.

## Example Resource State: Graph State

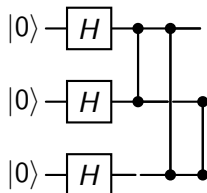
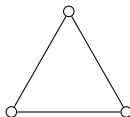
Let  $G = (V, E)$  be a graph. The **graph state**  $|G\rangle$  has

- ▶ For every  $v \in V$ , prepare a qubit  $|+\rangle = H|0\rangle$ .
- ▶ For every  $e = (u, v) \in E$ , apply  $CZ$  on qubits of  $u$  and  $v$ .

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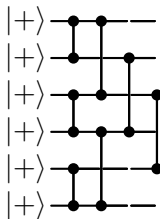
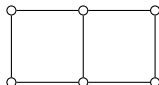
# Grid Graph State

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Example:  $|\text{Grid}_{2 \times 3}\rangle$ .



# Universal Resource, Informally

## Definition

Let LO denote local operations, i.e. single qubit unitaries and measurements. Let CC denote classical communication.

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## Definition

Let  $\Psi$  be a family of resource states.  $\Psi$  is a **universal resource** for MBQC if for every  $U \in SU(2^n)$ ,  $\forall n \in \mathbb{N}$ , there is a sequence of LOCC and some resource state  $|\psi\rangle \in \Psi$ , such that

$$|\psi\rangle \xrightarrow{\text{LOCC}} U|0^n\rangle$$

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In other words, we can “simulate any state” using LOCC and a resource state from  $\Psi$ .

# Grid is a Universal Resource

## Theorem

$\{\text{Grid}_{1 \times i}\}_{i \in \mathbb{N}}$  (The lines) are a 1-qubit universal resource for MBQC.

## Lemma (Euler decomposition)

Every single-qubit unitary can be expressed (up to global phase) as the product

$$R_z(\theta_3)R_x(\theta_2)R_z(\theta_1)$$

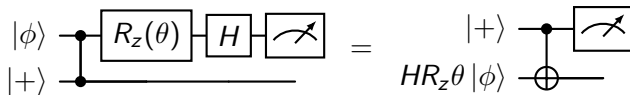
for some  $\theta_1, \theta_2, \theta_3 \in [0, 2\pi]$ .

## Lemma

$R_x(\theta) = HR_z(\theta)H$ , for all  $\theta \in [0, 2\pi]$ .

# Grid is a Universal Resource cont'd

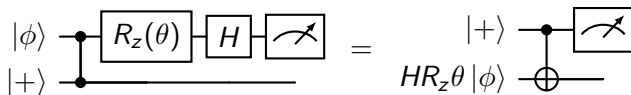
Informal proof.



□

# Grid is a Universal Resource cont'd

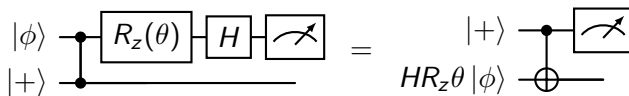
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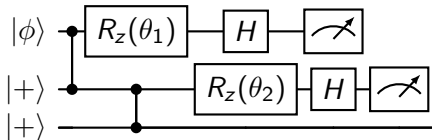
If we measure 0, we get  $HR_z(\theta)|\phi\rangle$  in the second register. If we measure 1, we get  $XH_z(\theta)|\phi\rangle$ . Thus given the right measurements we can apply any 1-qubit gate by repeating the construction:  $\square$

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# Grid is a Universal Resource cont'd

## Theorem

*Let Grid denote the family of all grid graphs. Grid is a universal resource for MBQC.*

## Proof.

Similar to the proof for the line.



# Comparability Grid

## Definition

Let  $\text{CGrid}_{m \times n}$  denote the  $m$  rows,  $n$  column **comparability grid**, which is an undirected graph  $G = (V, E)$  with

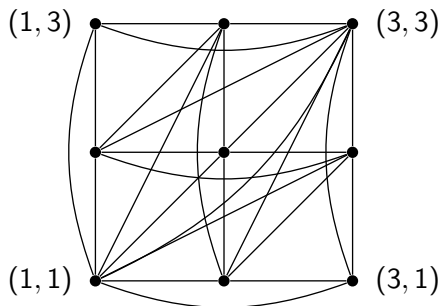
$$V = \{(i, j) \mid i \in [m], j \in [n]\}$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, i_2 \geq i_1, j_2 \geq j_1.\}$$

i.e. the grid with partial order  $\leq$  defining edges.

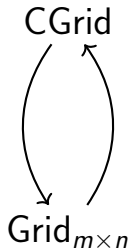
# Comparability Grid Example

Example:  $CGrid_{3 \times 3}$ :



# Proving Comparability Grid Universality

Strategy: Embed Grid in CGrid. Two ways to do this:

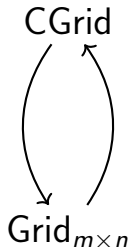


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## Method 1

What grids can we get  
from vertex deletion?

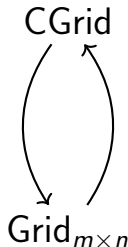


# Proving Comparability Grid Universality

Strategy: Embed Grid in CGrid. Two ways to do this:

## Method 1

What grids can we get  
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## Method 2

How can we embed  
Grid<sub>m×n</sub> in CGrid?

$\text{Grid}_{2 \times n}$  is a subgraph

### Lemma

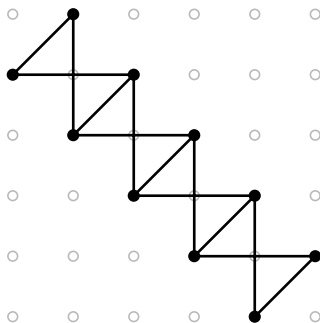
*For all  $n \in \mathbb{N}$ ,  $\text{Grid}_{2 \times n}$  is a subgraph of a sufficiently large CGrid.*

# $\text{Grid}_{2 \times n}$ is a subgraph

## Lemma

For all  $n \in \mathbb{N}$ ,  $\text{Grid}_{2 \times n}$  is a subgraph of a sufficiently large CGrid.

Proof.



# Which grids are subgraphs?

## Theorem

*Grid <sub>$k \times n$</sub>  is not a subgraph of any CGrid for  $k > 2$ ,  $n > 2$ .*

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## Proof.

Using a computer, we enumerate all possible 9 vertex subgraphs of CGrid. No such subgraph is isomorphic to Grid<sub>3×3</sub>. □

# Permutation Representation of Subgraphs

## Lemma

*Any subgraph  $H$  of  $CGrid$  with  $|V(H)| = n$  can be uniquely identified as a permutation of  $[n]$ , denoted  $\pi(H)$ .*

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## Proof.

Since  $H \subseteq \text{CGrid}$ , each vertex has distinct  $x, y$  coordinates. Name vertex  $i$  by its rank sorting by  $x + \frac{y}{n}$ . Denote  $\pi_i$  by the rank of vertex  $i$  sorted by  $y + \frac{x}{n}$ .

Conversely, given a permutation  $\pi \in S_n$ , construct vertices  $(i, \pi_i)$  for  $i = 1, \dots, n$ . This produces a subgraph of CGrid with  $n$  vertices.

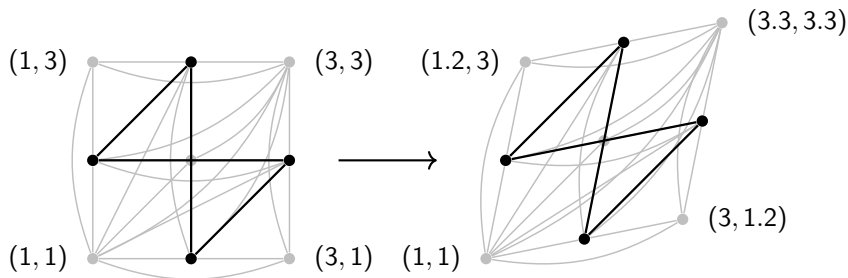
These are inverses of each other.



# Example Permutation Representation

## Example

The following subgraph  $H$  has  $\pi(H) = (2, 1, 4, 3)$ , since the leftmost vertex is 2nd lowest in  $y$  order, and so on.



# The Enumeration Algorithm

## Lemma

*Let  $H$  be a subgraph with  $|V(H)| = O(n)$  and  $|E(H)| = O(n)$ .  
Using a Fenwick tree, we can construct the degree sequence of  $H$   
from  $\pi(H)$  in two passes of  $O(n \log n)$  time.*

# The Enumeration Algorithm

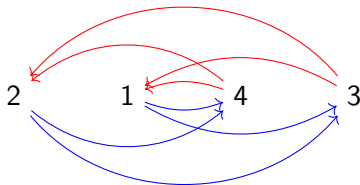
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In the **first pass**, construct the **backward degree**, i.e.  $\forall j < i$ , find  $\pi_i > \pi_j$ . In the **second pass**, run the algorithm on  $\pi(H)$  reversed to get the **forward degree**, i.e.  $\forall j > i$ , find  $\pi_i < \pi_j$ .

The sum of both is the degree sequence.

## Example



# The Enumeration Algorithm cont'd

## Lemma

Without even completing the *first pass*, we can identify if  $H$ ,  $|V(H)| = n$ , has more than  $c = O(n)$  edges. By setting a corresponding minimum, we can enumerate all  $H$  with  $|V(H)| \in [b, c]$  with  $O(n \log n)$  processing for each  $H$ .

## Example



# The Enumeration Algorithm cont'd

## Theorem

*Given an  $n \in \mathbb{N}$ , we can obtain a (small) set  $H$  with*

- ▶  $|V(H)| = n$
- ▶ *at least the number of edges in the target Grid*
- ▶ *at most  $k$  additional edges compared to the target Grid*
- ▶ *degree sequence at least the degree sequence of target Grid*
- ▶ *(some others)*

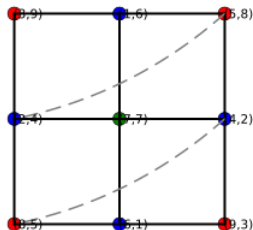
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# Recall: Which grids are subgraphs?

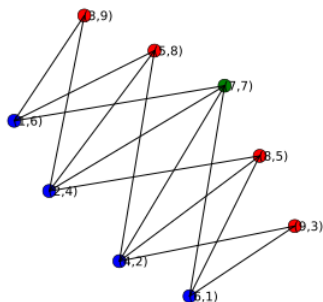
## Theorem

*$\text{Grid}_{k \times n}$  is not a subgraph of  $\text{CGrid}$  for  $k > 2$ . For  $k = 3$ , the minimum is 2 additional edges.*

Embedding 1



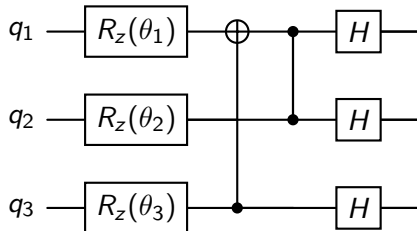
Comparability graph



# The Width-3 Gadget

## Definition

The previous graph gives rise to the following width-3 gadget:



Question: Is this gadget universal for MBQC?

# Lie Algebra Universality Algorithm

## Definition (Matrix Exponentiation)

For any  $A \in \mathbb{C}^{m \times m}$ ,

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}$$

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## Definition (Lie Algebra of $SU(2^n)$ )

$$\mathfrak{su}(2^n) = \{-iH \in \mathbb{C}^{2^n \times 2^n} \mid H^\dagger = H, \operatorname{tr}(H) = 0\}$$

# Lie Algebra Universality Algorithm cont'd

## Theorem

*A gate set  $\mathcal{G}$  is universal for  $n$  qubits if it generates  $SU(2^n)$ , the group of  $2^n \times 2^n$  unitary matrices with determinant 1.*

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*$U \in SU(2^n)$  if and only if  $U = e^H$  for some  $H \in \mathfrak{su}(2^n)$ .*

## Corollary

*To show  $\mathcal{G}$  is universal, it suffices to generate the Lie algebra.*

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## Fact

*$H \in \mathfrak{su}(2^n)$  if and only if  $H = -i \sum_{P \in \mathcal{P}_n \setminus \{I\}} \alpha_P P$  for some  $\alpha_P \in \mathbb{R}$ , where  $\mathcal{P}_n$  is the set of Pauli matrices on  $n$  qubits.*

## Corollary

*The traceless Hermitian operators are exactly those that can be represented as a linear combination of Pauli operators ( $X$ ,  $Y$ ,  $Z$ ).*

# Lie Algebra Universality Algorithm: Cliffords

## Lemma

*Denote by  $G(\theta_1, \theta_2, \theta_3)$  the width-3 gadget. Then  $G(0, 0, 0)$  is some fixed Clifford operation  $C$  and  $C \in \mathcal{G}$ .*

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## Lemma

*If  $C \in \mathcal{G}$  then  $C^\dagger \in \mathcal{G}$ .*

## Proof.

The Clifford group on 3 qubits is finite. Hence every element has finite order, i.e.  $\exists t$  with  $C^t = I$ . Thus  $C^\dagger = C^{t-1} \in \mathcal{G}$ . □

# Lie Algebra Universality Algorithm: Addition

## Fact (Baker-Campbell-Hausdorff formula)

For  $A, B \in \mathfrak{su}(2^n)$ ,

$$e^A e^B = e^{A+B+\frac{1}{2}[A,B]+\frac{1}{12}[A,[A,B]]+\frac{1}{12}[B,[A,B]]+\dots}$$

where  $[A, B] = AB - BA$  is the commutator of  $A$  and  $B$ .

## Fact

For  $A, B \in \mathfrak{su}(2^n)$ ,

$$\lim_{k \rightarrow \infty} e^{A/k} e^{B/k} = e^{A+B}$$

## Corollary

If we have  $A, B$  in the Lie algebra, we can generate their sum.

# Lie Algebra Universality Algorithm: Commutators

## Fact

For  $A, B \in \mathfrak{su}(2^n)$ ,

$$\lim_{k \rightarrow \infty} e^{B/k} e^{A/k} e^{-B/k} e^{-A/k} = e^{[A,B]}.$$

## Corollary

*If we have  $A, B$  in the Lie algebra, we can generate their commutator  $[A, B]$ .*

# Lie Algebra Universality Algorithm: Operations

Strategy: Start with some elements in the Lie algebra and generate more using:

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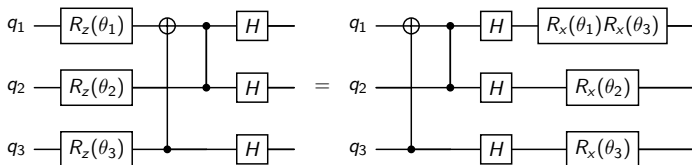
## Proof.

Brute force using a computer program.



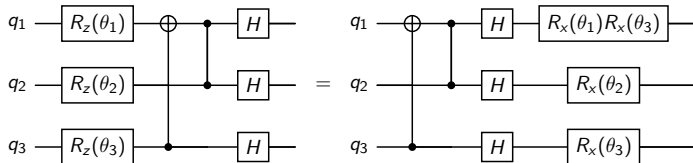
# Lie Algebra Universality Algorithm: Visual Intuition

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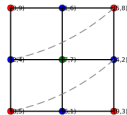
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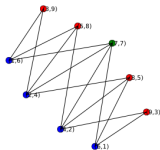
Idea: “Stitch” this with another gadget to “get”  $R_x(\theta_1)$  on  $q_1$ .

# Recall: Many width-3 gadgets.

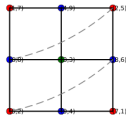
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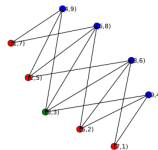
Comparability graph



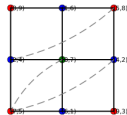
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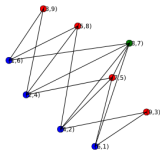
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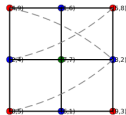
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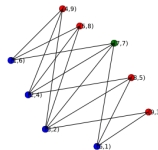
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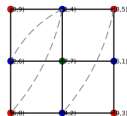
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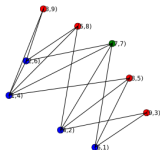
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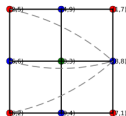
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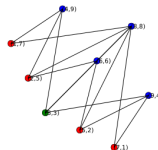
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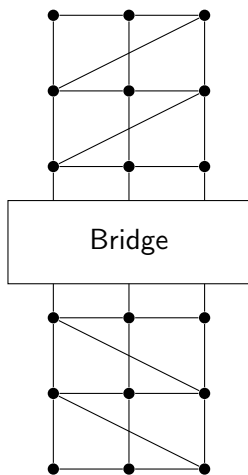
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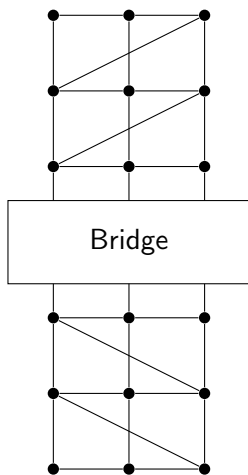
# Bridging Gadgets: Informal Visual Definition



## Definition (Informal)

A width-3 bridge of size  $b \in \mathbb{N} \cup \{0\}$  adds  $3b$  new vertices between two width-3 gadgets. The bridge has some  $\text{Grid}_{3 \times b}$  as a subgrid.

# Bridging Gadgets: Informal Visual Definition

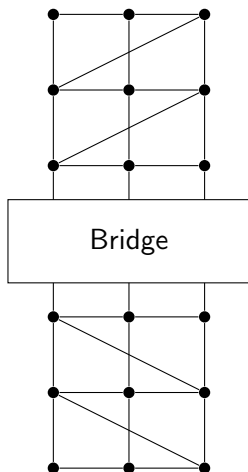


## Definition (Informal)

A width-3 bridge of size  $b \in \mathbb{N} \cup \{0\}$  adds  $3b$  new vertices between two width-3 gadgets. The bridge has some  $\text{Grid}_{3 \times b}$  as a subgrid.

The bridge can have any number of additional edges; set  $\theta = 0$  to “ignore” them.

# Bridging Gadgets: Informal Visual Definition



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The bridge can have any number of additional edges; set  $\theta = 0$  to “ignore” them.

Question: Given width-3 gadgets  $A$  and  $B$ , how can we construct a width-3 bridge of size  $b$  or prove that it is impossible?

# Linear Programming Bridging Algorithm

## Definition (Informal)

Given width-3 gadgets  $A$  and  $B$ , encode each as inequalities on a linear program.

## Example

Suppose we are dealing with width-2 gadgets and  $A$  has  $\pi(A) = (2, 1, 4, 3)$ . We add conditions:

$$x_1 < x_2 < x_3 < x_4$$

$$y_2 < y_1 < y_4 < y_3.$$

# Linear Programming Bridging Algorithm cont'd

## Lemma (Informal)

*We can add additional inequalities to a linear program encoding  $A$  to also encode  $B$  and any bridge vertices.*

## Theorem

*Fix a bridge size  $b$ . If the corresponding linear program is infeasible, then there is no bridge of size  $b$  between  $A$  and  $B$ .*

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## Fact

*There exists no bridge of size  $b < 4$  between any two width-3 gadgets  $A, B$ , where each gadget has at most 3 additional edges compared to  $\text{Grid}_{3 \times 3}$ .*

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## Proof.

Brute force every linear program encoding of ordered pairs of gadgets  $A, B$  with at most 3 additional edges. □

# Resolution?

## Conjecture

*CGrid is not a universal resource for MBQC.*

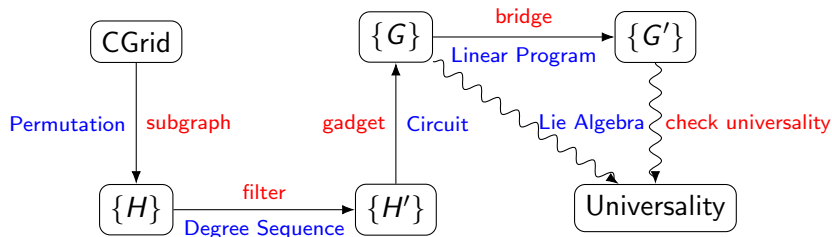
# Resolution?

## Conjecture

*CGrid is not a universal resource for MBQC.*

## Justification

*Considering gadgets with more additional edges is unlikely to help get  $R_z(\theta_1)$  on  $q_1$ .*



# Why care about comparability grids?

## Theorem (Van den Nest et al, 2006)

*Let  $\Psi$  be a family of resource states, and  $|\phi\rangle, |\phi'\rangle$  arbitrary states with the same number of qubits. Let  $E : |\phi\rangle \rightarrow \mathbb{R}$  be a function that satisfies the following properties:*

- 1.  $E(|\phi\rangle) \geq E(|\phi'\rangle)$  whenever  $|\phi\rangle \geq_{\text{LOCC}} |\phi'\rangle$ ;*
- 2.  $\sup_{|\phi\rangle} E(|\phi\rangle) > \sup_{|\psi\rangle \in \Psi} E(|\psi\rangle)$ .*

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*CGrids have unbounded rank width (and so do grid graphs).*

## Corollary

*If CGrid is not a universal resource, then unbounded rank width is not sufficient for universality.*